

Cilengitide affects tumor compartment, vascularization and microenvironment in experimental bone metastases as shown by longitudinal ^{18}F -FDG PET and gene expression analysis

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Abstract

Purpose Aim of this study was to investigate the specific treatment effects of inhibiting $\alpha v\beta 3/\alpha v\beta 5$ integrins by cilengitide in an animal model of breast cancer bone metastases using dynamic ^{18}F -FDG PET and gene expression analysis.

Methods For this purpose, nude rats bearing bone metastases were treated with cilengitide, a small molecule inhibitor of $\alpha v\beta 3$ and $\alpha v\beta 5$ integrins, from day 30 to 55 after tumor cell inoculation of MDA-MB-231 breast cancer cells (25 mg/kg, 5 days per week; $n = 8$ rats) and compared to control rats ($n = 8$). Dynamic ^{18}F -FDG PET data were assessed at days 30, 35 and 55 after tumor cell inoculation determining the vascular fraction VB and the metabolic variables k_1 – k_4 . At day 55, genome-wide mRNA expression analysis was performed to assess the

treatment-specific expression changes from cilengitide-treated and control rats.

Results In a longitudinal ^{18}F -FDG PET study, the vascular fraction VB was significantly decreased in bone metastases between days 30/35, 30/55 and 35/55, whereas the kinetic parameters k_1 and k_4 were significantly decreased between days 30/55 in skeletal lesions of treated animals. Gene expression analysis from bone metastases at day 55 revealed that tumor-produced integrins ($\alpha v\beta 5$) as well as factors relevant for angiogenesis ($\alpha v\beta 3$, VEGF, PDGF), bone resorption (PTHrP and RANKL), extracellular matrix remodeling (collagen, CD44) and bone marrow microenvironment (CXCR4) were significantly reduced upon therapy with cilengitide.

Conclusions Here, we provide evidence that cilengitide inhibits pivotal factors of all compartments of bone metastases including tumor cells, vasculature and bone microenvironment in vivo and by whole-genome transcriptome analysis.

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Keywords Bone metastases · Breast cancer · ^{18}F -FDG PET · Cilengitide · Integrins · Gene expression analysis

Introduction

In the pathogenesis of skeletal metastasis, the cross talk between tumor and stroma cells results in bone remodeling by osteoblasts and osteoclasts. Consequently, the microenvironment is enriched with bone-derived pathogenic factors that, in turn, enhance tumor signaling inducing progressed bone destruction. Novel therapies for metastatic disease in the skeleton target pivotal pathogenic factors such as RANKL (receptor activator of NF- κ B ligand) that stimulate tumor-induced bone resorption (Weilbaecher

et al. 2011). Also, the inhibition of integrins showed promising results for the treatment of bone metastases in experimental breast, prostate and lung cancer (Li et al. 2012; Nemeth et al. 2003; Pecheur et al. 2002; van der Horst et al. 2011; Zhao et al. 2007). Here, we examined the effect of cilengitide, an inhibitor of $\alpha v\beta 3$ and $\alpha v\beta 5$ integrins, in experimental breast cancer bone metastases on the tumor compartment, vascularization and microenvironment in these osseous lesions.

Integrins are key factors for tumor progression and metastasis formation. They are a family of 24 heterodimeric transmembrane glycoproteins that facilitate cell–cell and cell–extracellular matrix (ECM) interactions. Besides their role in recruiting many intracellular signaling molecules, integrins induce many important events for adhesion, migration, proliferation and apoptosis in the pathogenesis of metastasis (Desgrosellier and Cheresch 2010). The $\alpha v\beta 3$ and $\alpha v\beta 5$ integrins are strongly expressed on the tumor vasculature and activated endothelial cells, but also on tumor cells, including breast cancer cells (Max et al. 1997; Meyer et al. 1998). Furthermore, the overexpression of $\alpha v\beta 3$ integrin on breast cancer cells is associated with bone colonization of metastasized malignant cells and, consequently, osteoclast-mediated bone resorption (Cleazardin 2009).

The $\alpha v\beta 3$ and $\alpha v\beta 5$ integrins recognize the tripeptide sequences Arg-Gly-Asp (RGD) found in many extracellular ligands. Cilengitide (EMD 121974) is a cyclic pentapeptide containing the sequence RGD that selectively inhibits $\alpha v\beta 3$ and $\alpha v\beta 5$ integrins. Cilengitide acts on $\alpha v\beta 3/\alpha v\beta 5$ expressing cells, depriving them of signals needed for survival and proliferation, which resulted in the inhibition of angiogenesis and tumor growth (Beekman et al. 2006; Mas-Moruno et al. 2010; Reardon et al. 2011). However, conflicting discussions about pro- and anti-angiogenic effects of $\alpha v\beta 3/\alpha v\beta 5$ integrin inhibition were reported (Reynolds et al. 2009). Currently, cilengitide is in phase III trials for the treatment of glioblastoma multiforme and in phase II trials for many other solid tumors including lung and prostate cancer (Mas-Moruno et al. 2010). In experimental breast cancer bone metastases, cilengitide inhibited bone resorption and soft tissue tumor growth, modulated angiogenesis and showed an impact on ^{18}F -Fluorodeoxyglucose (FDG) kinetics using positron emission tomography (PET) (Bäuerle et al. 2011; Bretsch et al. 2011; Cheng et al. 2011).

The specific treatment effects of cilengitide, however, derived from longitudinal ^{18}F -FDG metabolism alterations and genetic expression patterns of tumor cells, vasculature and microenvironment have not been investigated. For this aim, ^{18}F -FDG kinetic parameters were determined by dynamic PET measurements to assess tumor and stroma metabolism in experimental breast cancer bone metastases during cilengitide treatment longitudinally. Further, gene expression profiling was performed to determine specific

genetic alterations upon $\alpha v\beta 3$ and $\alpha v\beta 5$ integrin inhibition by cilengitide in these skeletal lesions.

Materials and methods

Cell lines and culture conditions

The human estrogen-independent breast cancer cell line MDA-MB-231 was obtained from American type culture collection (ATCC). MDA-MB-231 cells were cultured routinely in RPMI-1640 (Invitrogen, Karlsruhe, Germany), supplemented with 10 % FCS (Sigma, Taufkirchen, Germany). All cultures were kept under controlled conditions (37 °C, 5 % CO_2 , humidified atmosphere) and passaged 2–3 times a week to keep them in logarithmic growth.

Animal model and study groups

Experiments performed in this study were approved by the responsible governmental animal ethics committee (Regierungspräsidium Karlsruhe, Germany). Nude rats (RNU strain; Harlan-Winkelmann GmbH, Borcheln, Germany) at the age of 6 weeks were housed in a specific pathogen-free environment in a mini barrier system under controlled conditions. For all procedures including tumor cell inoculation and in vivo imaging, rats were anesthetized using a mixture of nitrous oxide (1 l/min), oxygen (0.5 l/min) and isoflurane (1.5 vol%).

Animal model

The in vivo animal model has been described elsewhere and is summarized briefly (Bäuerle et al. 2005). A clear cut of 2–3 cm length was performed in the inguinal region. After preparation of all arterial branches, the flow of the femoral artery (FA) was temporarily occluded by clips that were placed proximal and distal of the superficial epigastric artery's (SEA) origin. In addition, the SEA was ligated distally and after making an incision proximal of the ligation, a 1 % papaverine solution was administered onto the SEA to facilitate the subsequent insertion of a needle (0.3 mm diameter and 42 mm length). After insertion, the needle was fixed in an external support. MDA-MB-231 cells (10^5 cells suspended in 0.2 ml media) were slowly injected into the SEA. Resulting bone metastases were observed exclusively in the femur, tibia and fibula of the right hind leg of these animals.

Study groups

A total of 20 nude rats were inoculated with MDA-MB-231 cells. Thirty days after cancer cell inoculation, nude rats

bearing bone metastases ($n = 16$) were randomly distributed into two groups. One group received the cyclic RGD peptide inhibitor of $\alpha v\beta 3/\alpha v\beta 5$ integrins (cilengitide, EMD 12197414; Merck, Darmstadt, Germany) intraperitoneally five times per week in isotonic saline (25 mg/kg; $n = 8$ rats), and the other group ($n = 8$ rats) served as sham-treated control. The observation period of all animals was 55 days, whereas 3 rats in the study died during PET imaging at day 35 (2 treated and 1 control rat) and 3 control animals had to be killed due to ethical reasons at day 45. As a result, the following numbers of animals were examined throughout the observation period: day 30, $n = 8$ treated and $n = 8$ control rats; day 35, $n = 6$ treated and $n = 7$ control rats; day 55, $n = 6$ treated and $n = 4$ control rats. In a longitudinal study, rats were imaged at day 30, 35 and 55 using PET as described below.

Positron emission tomography

After the inoculation of cancer cells, each rat was imaged at days 30, 35 and 55 using a dedicated PET system (ECAT EXACT HR+; Siemens, Erlangen, Germany) with an axial field of view of 15.3 cm operated in 2D mode. All animals were fastened for at least 4 h. The system provides the simultaneous acquisition of 63 transverse slices with a slice thickness of 2.4 mm. The animals were positioned in the axial plane of the system to maintain the best resolution in the center of the system. For all procedures, rats were anesthetized using a mixture of nitrous oxide (1 l/min), oxygen (0.5 l/min) and isoflurane (1.5 vol%). The standard ^{18}F -FDG PET protocol at our institute is based on a dynamic data acquisition protocol for 60 min, beginning with the intravenous injection of 1.80–5.67 MBq ^{18}F -deoxyglucose (^{18}F -FDG). The protocol comprises 28 frames (ten frames of 30 s, five frames of 60 s, five frames of 120 s and eight frames of 300 s; Strauss et al. 2008, 2009). A transmission scan (10 min) was obtained prior to radionuclide administration for attenuation correction of the acquired dynamic emission data. The software developed at our center was used to reconstruct images iteratively using the ordered-subset expectation maximization algorithm (6 iterations, 2 subsets), and the median root prior correction was applied to the intermediate matrices, which limits the dependency of the pixel maximum from the number of iteration steps. The theoretic voxel size was $2.277 \times 2.277 \times 2.425$ mm. The reconstructed PET images were automatically converted to SUV images.

The quantitative evaluation of the PET data was based on time-activity curves retrieved from volumes-of-interest (VOIs). One VOI was placed over the aorta to obtain the blood concentration of the tracer. Other VOIs were positioned over the tumor and normal tissue. Partial volume correction was applied according to the size of the VOIs.

The recovery is 0.85 for a diameter of 8 mm and 0.32 for a diameter of 3 mm. The standard model for FDG is a two-tissue compartment model, providing the following parameters: the transport parameters for ^{18}F -FDG into and out of the cell (k_1 and k_2), the parameters for phosphorylation and dephosphorylation of the intracellular ^{18}F -FDG (k_3 and k_4) and the fractional blood volume (VB, also called vessel density) that reflects the amount of blood in the volume of interest (Pan et al. 2007; Strauss et al. 2008, 2009; Cheng et al. 2011). A dedicated software developed at our center was used to obtain the compartment parameters. The software is based on a modified support vector machines algorithm (SVM) and a data base of compartment data (Pan et al. 2007). The modified SVM algorithm was applied to obtain an estimate of the compartment parameters. Then, the classical Levenberg–Marquardt algorithm could be used to optimize the model fitting. Curve fitting was based on the total X^2 difference as the cost function, where the primary criterion was to minimize the summed squares of the differences for the measured and model curve. Usually, X^2 was less than two. The model parameters were acceptable for a range of 0–1.

Statistical analysis

For statistical comparisons, all kinetic PET parameters and SUV, using a single parameter analysis, were compared between the control and treatment groups using two-sided Wilcoxon test; p values ≤ 0.05 were considered significant.

Histology

At day 55 after tumor cell inoculation, animals were sacrificed. Lower limbs of control animals were excised and processed as described previously (Bäuerle et al. 2010). For immunostaining, the sections (embedded in Technovit® 9100 NEU, Heraeus Kulzer, Germany) were incubated overnight with the following primary antibodies: rabbit anti-collagen IV pAB (1:50, Progen Biotechnik GmbH, Heidelberg, Germany), mouse anti-smooth muscle actin (SMA) polyclonal antibody (1:400; Sigma Aldrich, Saint Louis, MO, USA), goat anti-EGFR polyclonal antibody (1:100, Santa Cruz Biotechnology, Inc.), goat anti-RANKL polyclonal antibody (1:100, Santa Cruz Biotechnology Inc., Heidelberg, Germany), rabbit anti-EGF polyclonal antibody (1:100, Novus Biologicals, USA) and mouse anti-CXCR4 polyclonal antibody (1:100, R&D Systems GmbH, Wiesbaden, Germany). Sections were incubated with the secondary antibody for 1 h at room temperature as follows: Alexa Fluor® 488-conjugated donkey anti-rabbit IgG, Alexa Fluor® 594-conjugated donkey anti-goat IgG and DyLight™ 594-conjugated goat anti-mouse IgG

(1:100, respectively, Jackson ImmunoResearch, Suffolk, UK). After a nuclear staining step with DAPI (4',6-diamidino-2-phenylindole, Serva, Germany), sections were mounted in Fluoromount G (Southern Biotech, USA). Sections were examined using a Leica microscope (DMRE Bensheim, Germany) equipped with an adapted digital camera.

Gene expression analysis

RNA extraction

Total RNA was extracted from experimental bone metastases after the observation time from $n = 5$ rats of the therapy group and $n = 3$ control animals using the RNeasy Mini kit (Qiagen, Hilden, Germany). Approximately 30 mg tissue was homogenized with 600 μ l Buffer RLT (β -ME added; Qiagen) and a QIAshredder (Qiagen). Lysate was centrifuged for 3 min at maximum speed in a microcentrifuge, and the supernatant was transferred into a new tube. Next, 1 volume (600 μ l) of 70 % ethanol was added to the cleared lysate. 700 μ l of the sample was applied to an RNeasy mini spin column in a 2 ml collection tube and centrifuged for 15 s at maximum speed. The column was washed with 700 μ l RW1 (Qiagen) and 500 μ l RPE (Qiagen) (loaded and washed through by centrifugation for 15 s at maximum speed). An additional 500 μ l RPE was loaded on the column and washed by centrifugation for 2 min at maximum speed. Finally, the RNA was eluted from the column by applying 50 μ l RNase-free H₂O and centrifuging for 1 min at maximum speed. Concentration and quality were determined with Agilent 2100 Bioanalyzer (Agilent Technologies, USA). Only samples with a RNA integrity number (RIN) >5.0 and no evidence of ribosomal degradation were included in the study.

Expression analysis

Expression analysis was performed by the DKFZ Genomic and Proteomics core facility following manufactures instructions. Briefly, biotin-labeled cRNA is added to Illumina HumanHT-12 v4 BeadChip (Illumina, USA) that analyzes 47,000 genes on one array. Raw values were log2 transformed. Between human and rat RNA, no considerable cross-species detection is expected as reported recently for human and mouse samples using comparable expression beadchips (Iorns et al. 2012). Subsequently, we preselected the top 500 genes, which showed the most significant expression change (Wilcoxon test, $p < 0.01$) between the treatment group and control animals. To investigate treatment-specific signaling pathway alterations,

we applied Ingenuity Pathways Analysis (IPA; Ingenuity Systems, <http://www.ingenuity.com/>). Here, the rank-product-generated gene lists cut at 50 % false discovery rate were uploaded into the IPA server as input data. IPA uses pathway libraries derived from the scientific literature. Significance of the biofunctions was tested by Fisher's exact test.

Results

Dynamic ¹⁸F-FDG PET parameters

FDG uptake (standard uptake value, SUV) was significantly decreased in animals treated with cilengitide between day 30/55 ($p < 0.01$) and 35/55 ($p < 0.05$), but not between the groups at any time point. Representative images of FDG uptake are shown in Fig. 1. Determination of kinetic parameters indicated that the vascular fraction VB was significantly decreased in bone metastases between days 30/35 (Fig. 2a; $p \leq 0.05$), 30/55 (Fig. 2b; $p < 0.05$) and 35/55 (Fig. 2c; $p < 0.05$). The kinetic parameters k1 and k4 were significantly decreased between days 30/55 (Fig. 2b; $p < 0.05$) and parameter k1 between days 35/55 (Fig. 2c; $p < 0.05$) in skeletal lesions of treated animals. In control rats, the parameters VB, k2 and k3 were significantly decreased between days 35/55 (Fig. 2f; $p < 0.05$). When comparing between the groups, the vascular fraction VB and transport rate k1 were significantly higher in controls at day 55 (VB, $p < 0.01$; k1, $p < 0.05$).

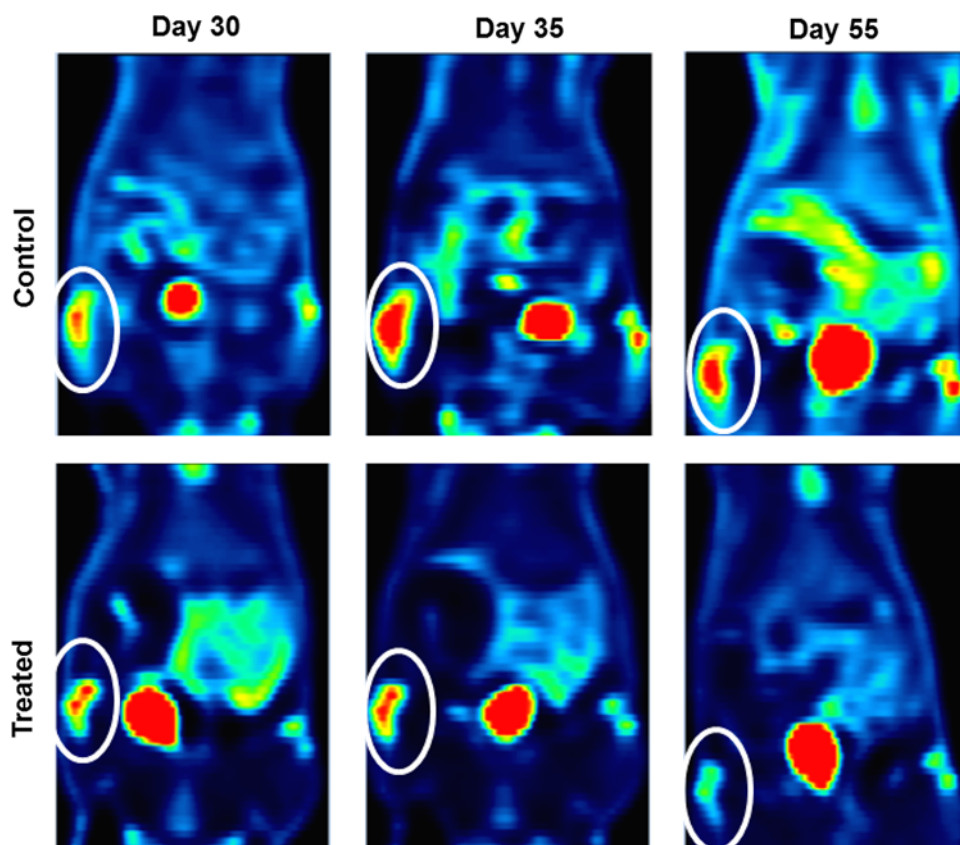
Gene expression analysis

In order to assess the specific effects of cilengitide ex vivo in experimental breast cancer bone metastases, the expression pattern was studied using Illumina whole-genome expression arrays. For this purpose, the top 500 genes that were expressed significantly different ($p < 0.01$) between the cilengitide-treated and control groups were included. Some examples of factors produced by tumor cells in the vascular compartment and within the micro-environment are referred to below and listed in Table 1.

Tumor cells

Integrin subunits α_v , β_3 and β_5 that are expressed by MDA-MB-231 breast cancer cells were significantly downregulated in animals treated with cilengitide (Table 1). The mRNA content of intracellular hexokinase 1 (HK1) or hexokinase 3 (HK3) as well as glucose transporter 1 (GLUT1) was also significantly decreased in cilengitide-treated animals compared to controls (Table 1).

Fig. 1 Representative SUV images of ^{18}F -FDG uptake in axial orientation from treated and untreated animals at days 30, 35 and 55 after cancer cell injection (see the *right* hind leg, circles delineate bone metastases)



Vascular compartment

Treatment with cilengitide resulted in significantly down-regulated angiogenic growth factors including VEGF A, PDGF B and angiopoietin 2 in comparison with untreated animals (Table 1). Genes involved in angiogenic pathway signaling, such as TP53/11 (tumor protein p53 inducible protein 11) and ROBO2 (roundabout homolog 2), were also significantly downregulated in treated animals compared to controls (Table 1). In addition, gene types of cell adhesion (cadherins and claudins) and vessel wall components (collagen IV) were differently expressed between control and treated animals (Table 1).

Microenvironment

Comparison of treated and control animals resulted in significant differences concerning the expression of factors influencing osteoclast and osteoblast function. Components that contribute to osteoclast function including PTHrP (parathyroid hormone-related protein), RANKL, IL1 (interleukin 1), IL6, IL10, EGF (epidermal growth factor) and CSFs (colony-stimulating factors) were significantly downregulated in cilengitide-treated animals (Table 1). On the contrary, osteoblast-modulating factors including TGF β , BMP2 (bone morphogenetic protein 2), TNF α

(tumor necrosis factor alpha) were significantly upregulated after treatment (Table 1). Controls showed a significantly higher expression of CXCR4 in comparison with therapy group (Table 1). Furthermore, significant down-regulation of genes associated with ECM remodeling (ADAM and CD44) was observed after cilengitide (Table 1).

Biological function

Functional annotation of the top 500 genes derived from Ingenuity software indicated that cellular movement, growth and proliferation were among the functions that differed significantly between the groups (Table 2). Further processes including cell death and cell-to-cell signaling/interaction were also significantly different (Table 2).

Histology

Some of the above-mentioned differently expressed factors could be localized in bone metastases by immunohistochemistry: RANKL, CXCR4, EGF, EGFR. Thereby, RANKL was found to be localized at the tumor/bone interface of skeletal metastases (Fig. 3a, a'), whereas CXCR4 was more diffusely distributed among the tumor cells (Fig. 3b, b'). EGF and EGFR, however, were

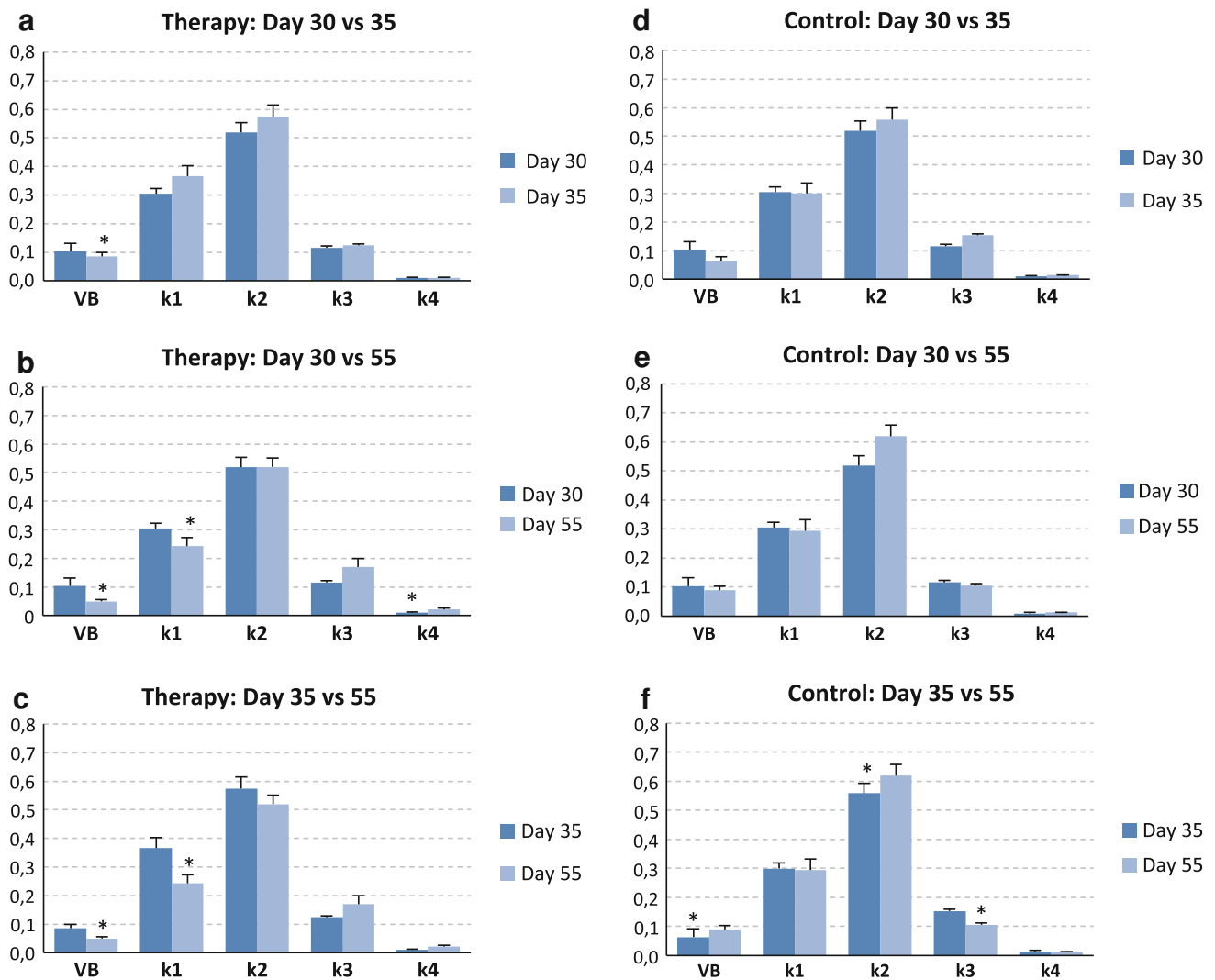


Fig. 2 a–f PET parameters for the FDG kinetics in treated (left) and untreated (right) animals expressed as $\text{ml} \times \text{ml}^{-1}$ (vB) $\text{ml} \times \text{ml}^{-1}$ tissue $^{-1} \times \text{min}^{-1}$ (k1); l/min (k2, k3, k4) of the longitudinal study.

Error bars indicate standard error. Asterisks denote a significant difference by the two-sided Wilcoxon test between treatment and control groups (* $p \leq 0.05$; ** $p < 0.01$)

expressed regionally by fractions of MDA-MB-231 cells only (Fig. 3c, c', d, d'). SMA and collagen IV stained for the vascular structures pericytes and basement membrane, respectively, showing a tortuous and irregular vascular network particularly at the rim of bone metastases (Fig. 3a–d).

Discussion

Integrin inhibition was shown to be a promising approach for the treatment of metastasis to bone induced by breast, prostate and lung cancer. The aim of this study was to assess specific effects of $\alpha\nu\beta3/\alpha\nu\beta5$ integrin inhibition by cilengitide on breast cancer bone metastases in nude rats by examining ^{18}F -FDG metabolic parameters longitudinally

and to analyze gene expression changes of experimental bone metastases after treatment. We thereby found significant alterations upon cilengitide treatment affecting all compartments of bone metastases including cancer cells, vasculature and microenvironment.

Between ^{18}F -FDG kinetics assessed by PET and tumor angiogenesis, a correlation is anticipated, as angiogenesis determines the vessel density of a tumor, but it also influences glucose transporter (Haberkorn et al. 2007; Strauss et al. 2008). Cilengitide decreased the vascular fraction VB (also called vessel density) in bone metastases already after 5 days of treatment and over the complete observation time. These findings are in good agreement with previous studies, which described an inhibition of regional blood volume determined by dynamic contrast-enhanced MRI (DCE-MRI) during cilengitide treatment

Table 1 Gene expression analysis

Gene	Description	Function (incomplete list)	Log2
ADAM10	ADAM metallopeptidase domain 10	ECM remodeling	−0.14
ADAM29	ADAM metallopeptidase domain 29	ECM remodeling	−0.54
ADAM3A	ADAM metallopeptidase domain 3A	ECM remodeling	−0.41
ADAM5	ADAM metallopeptidase domain 5	ECM remodeling	−0.83
ANGPT2	Angiopoietin 2	Angiogenesis	−0.45
BMP2	Bone morphogenetic protein 2	Osteoblastic activity	0.28
CD44	CD44 molecule	Cell adhesion	−0.25
CDH11	Cadherin 11	Endothelial adherens junctions	−0.4
CLDN18	Claudin 18	Endothelial tight junctions	−0.64
CLDN22	Claudin 22	Endothelial tight junctions	−0.45
COL4	Collagen, type IV	ECM remodeling	−0.11
COL6	Collagen, type VI	ECM remodeling	−0.42
CSF1	Colony-stimulating factor 1	Osteolysis	−0.39
CSF2	Colony-stimulating factor 2	Osteolysis	−0.51
CXCR4	Chemokine (C-X-C motif) receptor 4	Osteolysis, progression metast. breast cancer	−0.64
EGF	Epidermal growth factor	Osteolysis/osteoblastic activity	−0.33
EGFR	Epidermal growth factor receptor	Progression metastatic breast cancer	0.14
FGF5	Fibroblast growth factor 5	Osteolysis	−0.42
GLUT1	Glucose transporter 1	Glucose transport	−0.34
HIF1AN	Hypoxia inducible factor 1 alpha	Angiogenesis	0.38
HK1	Hexokinase 1	Glucose transport	−0.14
HK3	Hexokinase 3	Glucose transport	−0.34
IGF	Insulin-like growth factor	Osteolysis/osteoblastic activity	−0.59
IGF1	Insulin growth factor 1	Osteolysis/osteoblastic activity	−0.37
IL1	Interleukin 1	Osteolysis/osteoblastic activity	−0.4
IL2	Interleukin 2	Progression metastatic breast cancer	−0.63
IL6	Interleukin 6	Osteolysis	−0.52
IL10	Interleukin 10	Progression metastatic breast cancer	−0.51
ITGA5	Integrin alpha 5	Angiogenesis	−0.35
ITGAV	Integrin alpha v	Cell adhesion, angiogenesis	−0.55
ITGB3	Integrin beta 3	Cell adhesion, angiogenesis, osteolysis	−0.64
ITGB5	Integrin beta 5	Cell adhesion, angiogenesis	−0.4
MMP11	Matrix metalloproteinase 11	ECM remodeling	0.41
MMP14	Matrix metalloproteinase 14	ECM remodeling	0.25
MMP26	Matrix metalloproteinase 26	ECM remodeling	−0.45
PDGFB	Platelet-derived growth factor beta	Angiogenesis	−0.33
PTHrP	Parathyroid hormone-related peptide	Osteolysis/osteoblastic metastasis	−0.73
RANKL	Receptor activator of NF-κB ligand	Osteolysis	−1.28
ROBO2	Roundabout homolog 1	Angiogenesis	−0.41
TGFB3	Transforming growth factor beta 3	Osteolysis/osteoblastic activity	0.41
TNFA	Tumor necrosis factor alpha	Osteolysis/osteoblastic activity	0.54
TP53I11	Tumor protein p53 inducible protein 11	Angiogenic signaling	−0.66
VEGFA	Vascular growth factor alpha	Angiogenesis	−0.33

Genes produced by tumor cells, the vascular compartment and the microenvironment in experimental breast cancer bone metastasis among the top 500 that were expressed significantly different between cilengitide-treated and control groups ($p < 0.01$; incomplete list)

(Bäuerle et al. 2011; Bretschgi et al. 2011). Accordingly, our study showed downregulation of angiogenesis-related genes, particularly VEGF A and angiopoietin 2 in treated

animals as assessed by gene-array technique. Several studies reported that $\alpha v\beta 3$ can directly regulate the production of VEGF, and blocking $\alpha v\beta 3$ function with the

Table 2 Biological functions

Biological functions	<i>p</i> value
Cellular movement	0.00002
Cellular growth and proliferation	0.0001
Cell death	0.0004
Cell-to-cell signaling and interaction	0.0004
Cellular assembly and organization	0.0004
Molecular transport	0.001
Cellular development	0.003

Treatment induced signaling alterations of biological functions as performed by ingenuity pathway analysis. Significance of the given functions was tested by the Fisher's exact test (*p* values <0.05 were considered significant)

cyclic RGD peptide in a model of ischemic stroke reduced the production of VEGF (De et al. 2005; Shimamura et al. 2006). In line with our results, previous reports described a correlation between transport constant *k*₁ and expression of VEGF and angiopoietin 2 (Strauss et al. 2008). Furthermore, an experimental study revealed that ¹⁸F-FDG uptake was significantly correlated with the expression of GLUT1, which was again co-regulated with VEGF (Pedersen et al. 2001).

The parameters *k*₁ and *k*₄, which reflect the transport of ¹⁸F-FDG into the cell and the dephosphorylation of intracellular ¹⁸F-FDG, respectively, were significantly decreased in the later stages of treatment when compared to pre-treatment values. Gene expression data showed significantly lower expression values of GLUT1 and hexokinases 1 and 3 in cilengitide-treated animals. This is in line with previous studies demonstrating that *k*₁ is associated with glucose transporters and most sensitive for therapeutic changes (Brown et al. 2002; Strauss et al. 2008; Cheng et al. 2011). These findings indicated an effect of cilengitide treatment on the global ¹⁸F-FDG metabolism, and thus on the tumor cell compartment of bone metastases suggesting that cilengitide affected both, initially the vascular compartment and subsequently tumor cell metabolism. Metabolic tumor cell changes might therefore be a consequence of the anti-angiogenic effects of cilengitide.

As reported previously, the inhibition of $\alpha v\beta 3/\alpha v\beta 5$ integrins in experimental breast cancer bone metastases by cilengitide resulted in vessel remodeling including smaller and partly non-functional blood vessels with increased permeability (Bäuerle et al. 2011). This altered vessel phenotype might be responsible for the limited transport of ¹⁸F-FDG into the cell as assessed by parameter *k*₁ in the current study after cilengitide treatment. In addition, we found downregulated adhesion molecules such as claudins, cadherins and CD44 after inhibition of $\alpha v\beta 3/\alpha v\beta 5$ integrins. These molecules have a key role in epithelial cell–cell adhesion, and cilengitide may elicit signaling events

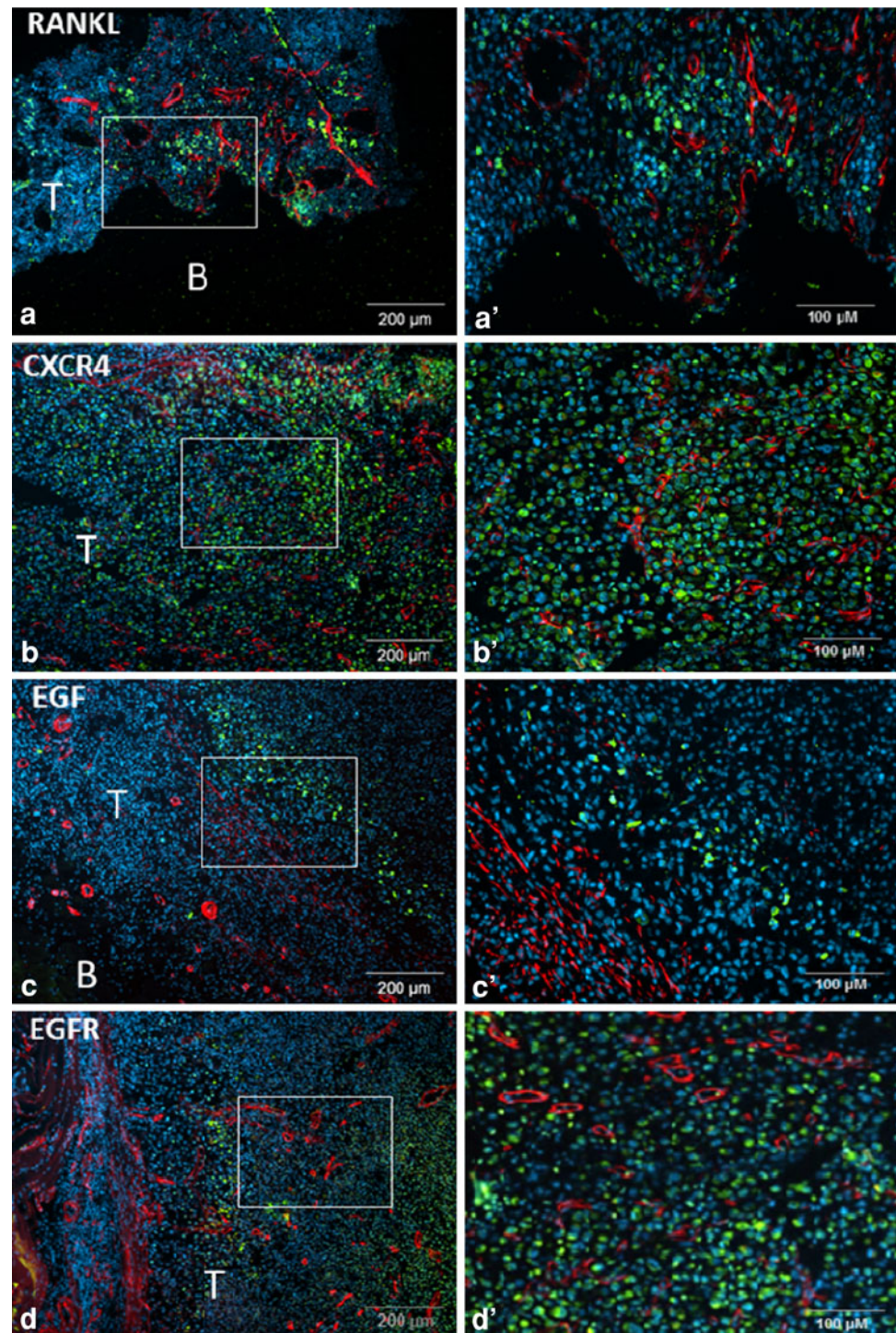
disrupting adherens junctions (Alghisi et al. 2009). Increased vessel permeability after cilengitide treatment may thus be a consequence of the loss of cellular contacts due to a lack of cadherin and claudin expression. Therapeutic administration of cyclic RGD peptides was also previously reported to disrupt cell–matrix adhesion of integrins increasing endothelial permeability in bovine lungs (Curtis et al. 1995).

In the microenvironment of bone metastases, the cross talk between tumor and bone or bone marrow cells results in tumor cell proliferation and alteration of bone metabolism (Mundy 2002). In this environment, several factors that are produced by breast cancer cells or the microenvironment including PTHrP, RANKL and interleukins stimulate osteoclast activity (Mundy 2002). Here, we observed significantly decreased gene expression values for such factors including PTHrP, RANKL, IL1, IL6 and TNF- α in cilengitide-treated animals, which directly or indirectly stimulate osteoclast maturation and activity (Clezardin 2011). Furthermore, we found downregulated expression of other bone matrix proteins that are released during osteoclastic resorption, for example, IGF1 and PDGF B upon administration of cilengitide. Our observations are in good agreement with previous findings from experimental breast cancer bone metastases showing decreased bone resorption after cilengitide treatment (Bäuerle et al. 2011; Bretsch et al. 2011). Interestingly, in a previous study, animals treated with cilengitide were reported to present with new bone formation within osteolytic lesions (Bäuerle et al. 2011). Accordingly, we found upregulation of osteoblast stimulating factors including TGF β , BMP2, TNF α after treatment with cilengitide. Taken together, these findings suggest an impact of cilengitide on pivotal factors in the bone microenvironment of metastatic breast cancer that particularly affect osteoclastic and osteoblastic function in the pathogenesis of skeletal lesions.

A limitation of our study is the relatively small number of animals enclosed, requiring larger studies for confirmation of the presented results. With an increased number of available tissue specimen after the dynamic PET scan, significant correlations between PET parameters and expressed genes might be determined (Strauss et al. 2008).

Current standard treatments for breast cancer bone metastases are limited to hormone therapy, chemotherapy and bisphosphonates. Currently, several targeted therapies are being investigated clinically for its use in bone metastatic breast cancer including anti-angiogenic drugs and denosumab, a monoclonal antibody against RANKL that inhibits the function of osteoclasts (Brown and Coleman 2011). All these treatment approaches, however, primarily interact with one compartment of bone metastases—either the tumor cell, vascular or bone compartment. Cilengitide, in contrast, targets all compartments of bone metastases:

Fig. 3 a–d Histologic analysis of experimental breast cancer bone metastases of control rats. **a** Immunofluorescence (IF) against RANKL (green) counterstained with collagen IV (red) and DAPI (blue). **b** Green color shows CXCR4 staining, whereas red denotes collagen IV; blue cell nuclei (DAPI). **c** IF against EGF (green) and smooth muscle actin (red), blue DAPI. **d** Green color shows EGFR counterstained with collagen IV (red) and DAPI (blue). Enlarged images of the indicated structures (left, white rectangle) are presented on the right side (**a'**, **b'**, **c'**, **d'**). “T” indicates tumor and “B” bone tissue of skeletal metastases. **a–d** Bar 200 μ m; **a'–d'** bar 100 μ m



tumor cells, vasculature and bone microenvironment. Due to the favorable safety profile of cilengitide compared to currently used treatments, this $\alpha v\beta 3/\alpha v\beta 5$ integrin inhibitor is a promising novel therapy for breast cancer metastasis to bone, which should be evaluated further in clinical studies.

Here, we report for the first time the metabolic changes of experimental breast cancer bone metastases in a longitudinal dynamic PET study and the resulting impact on gene expression upon treatment with cilengitide. Thereby, we demonstrate that cilengitide affects all compartments of bone metastases including metastatic cancer cells, tumor

vasculature and bone marrow microenvironment. Although confirmatory data are necessary, cilengitide is a promising treatment option for metastases in the skeleton by targeting all components of these osseous lesions.

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Conflict of interest The authors declare that they have no conflict of interest.

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